

```
1 C:\Users\Aisha\AppData\Local\Programs\Python\
  Python312\python.exe C:\Users\Aisha\PycharmProjects\
  PythonProject8\final_exp_pipeline2.py
2 =====
  =====
3 STEP 1: LOADING SPML DATA
4 =====
  =====
5 Total rows: 16012
6 Label distribution:
7 label
8 1      12542
9 0      3470
10 Name: count, dtype: int64
11
12 =====
   =====
13 STEP 2: BALANCING DATASET
14 =====
   =====
15 Balanced: 6940 prompts
16   Attacks: 3470
17   Innocent: 3470
18
19 =====
   =====
20 STEP 3: SPLITTING 80/20
21 =====
   =====
22 Train: 5552 | Test: 1388
23
24 =====
   =====
25 EXPERIMENT A: MiniLM + SVM
26 =====
   =====
27
28 Loading MiniLM...
29 Warning: You are sending unauthenticated requests to
   the HF Hub. Please set a HF_TOKEN to enable higher
   rate limits and faster downloads.
```

```
30 Loading weights: 100%|██████████| 199/199 [00:00<00:
    00, 15804.49it/s]
31 Model loaded!
32
33 Encoding training user prompts...
34 Batches: 100%|██████████| 174/174 [01:30<00:00, 1.
    93it/s]
35
36 Encoding test user prompts...
37 Batches: 100%|██████████| 44/44 [00:23<00:00, 1.84it
    /s]
38
39 Training SVM...
40 SVM training complete!
41
42 EXPERIMENT A RESULTS (MiniLM + SVM):
43   Accuracy:   0.9892
44   Precision:  0.9985
45   Recall:     0.9798
46   F1:         0.9891
47   FPR:        0.0014
48   FNR:        0.0202
49
50 Confusion Matrix:
51   True Positives: 680
52   True Negatives: 693
53   False Positives: 1
54   False Negatives: 14
55
56 =====
    =====
57 EXPERIMENT B: ZERO-SHOT NLI
58 cross-encoder/nli-MiniLM2-L6-H768
59 No training – direct classification
60 =====
    =====
61
62 Loading zero-shot NLI pipeline...
63 Loading weights: 100%|██████████| 106/106 [00:00<00:
    00, 3822.05it/s]
64 NLI model loaded!
```

```

65
66 Running zero-shot NLI on 1388 test prompts...
67 (This may take several minutes on CPU)
68   Processing 0/1388...
69   Processing 100/1388...
70   Processing 200/1388...
71   Processing 300/1388...
72   Processing 400/1388...
73   Processing 500/1388...
74   Processing 600/1388...
75   Processing 700/1388...
76   Processing 800/1388...
77   Processing 900/1388...
78   Processing 1000/1388...
79   Processing 1100/1388...
80   Processing 1200/1388...
81   Processing 1300/1388...
82
83 EXPERIMENT B RESULTS (Zero-shot NLI):
84   Accuracy:    0.5684
85   Precision:   1.0000
86   Recall:     0.1369
87   F1:         0.2408
88   FPR:        0.0000
89   FNR:        0.8631
90
91 Confusion Matrix:
92   True Positives: 95
93   True Negatives: 694
94   False Positives: 0
95   False Negatives: 599
96
97 =====
   =====
98 STEP 4: COMPARISON – MiniLM+SVM vs Zero-shot NLI
99 =====
   =====
100
101 Metric                               MiniLM+SVM   Zero-shot NLI
      Winner
102 -----

```

```
102 -----
103 Accuracy                0.9892          0.
    5684MiniLM+SVM
104 Precision                0.9985          1.
    0000Zero-shot NLI
105 Recall                  0.9798          0.
    1369MiniLM+SVM
106 F1                      0.9891          0.
    2408MiniLM+SVM
107 FPR                     0.0014          0.
    0000Zero-shot NLI
108 FNR                     0.0202          0.
    8631MiniLM+SVM
109
110 KEY FINDING:
111   MiniLM+SVM outperforms Zero-shot NLI by 42.07%
112   Training data provides 42.07% accuracy advantage
113   Zero-shot NLI insufficient for English injection
    detection
114
115 =====
    =====
116 STEP 5: SAVING RESULTS
117 =====
    =====
118   pipeline2_comparison.csv saved
119
120 =====
    =====
121 PIPELINE 2 COMPLETE
122 =====
    =====
123
124 Dataset:  SPML English (1388 test prompts)
125
126 MiniLM + SVM (trained):
127   Accuracy: 0.9892 | F1: 0.9891 | FPR: 0.0014
128
129 Zero-shot NLI (no training):
130   Accuracy: 0.5684 | F1: 0.2408 | FPR: 0.0000
131
```

132 Process finished with exit code 0

133